

1. This allows maximum flexibility for the migration process, and mirrors the naturally inspired quality that although two populations have the same physical separation, it may be easier to migrate in one direction than the other, i.e. fish migration is easier downstream than upstream. The migration of the *island* models is like the notion of migration in nature, being similar to the metapopulation models of theoretical ecology [26]. This model has also been used successfully in the determination of investment strategies in the commercial sector, in a product known as the Galapagos toolkit [45]. However, all the *islands* in this approach work on exactly the same problem, which makes it less analogous to biological ecosystems in which different locations can be environmentally different [3]. We will take advantage of this property later when defining the Ecosystem-Oriented Architecture of Digital Ecosystems.

5 The Digital Ecosystem

We will now define the architectural principles of Digital Ecosystems. We will use our understanding of theoretical biology [6, 4], mimicking the processes and structures of life, evolution, and ecology of biological ecosystems. We will achieve this by combining elements from mobile agents systems, distributed evolutionary computing, and Service-Oriented Architectures, to create a hybrid architecture which is the *digital counterpart of biological ecosystems*. Combining these technologies, based on the biological inspiration, the technical component of Digital Ecosystems provide a two-level optimisation scheme inspired by natural ecosystems, in which a decentralised peer-to-peer network forms an underlying tier of distributed agents. These agents then feed a second optimisation level based on an evolutionary algorithm that operates locally on single habitats (peers), aiming to find solutions that satisfy locally relevant constraints. The local search is sped up through this twofold process, providing better local optima as the distributed optimisation provides prior sampling of the search space by making use of computations already performed in other peers with similar constraints. So, the Digital Ecosystem supports the automatic combining of numerous agents (which represent services), by their interaction in evolving populations to meet user requests for applications, in a scalable architecture of distributed interconnected habitats. The sharing of agents between habitats ensures the system is scalable, while maintaining a high evolutionary specialisation for each user. The network of interconnected habitats is equivalent to the *abiotic* environment of biological ecosystems [3]; combined with the agents, the populations, the agent migration for distributed evolutionary computing, and the environmental selection pressures provided by the user base, then the union of the habitats creates the Digital Ecosystem, which is summarised in Figure 4. The continuous and varying user requests for applications provide a dynamic evolutionary pressure on the applications (agent aggregations), which have to evolve to better fulfil those user requests, and without which there would be no driving force to the evolutionary self-organisation of the Digital Ecosystem.

If we consider an example user base for the Digital Ecosystem, the use of SOAs in its definition means that business-to-business (B2B) interaction scenarios [22] lend themselves to being a potential user base for Digital Ecosystems. So, we can consider a *business ecosystem* of Small and Medium sized Enterprise (SME) networks [34], as a specific class of examples for B2B interaction scenarios; and in which the SME users are requesting and providing software services, represented as agents in the Digital Ecosystem, to fulfil the needs of their business processes, creating a Digital Business Ecosystem as shown in Figure 5. SOAs promise to provide potentially huge numbers of services that programmers can combine, via the standardised interfaces, to create increasingly more sophisticated and distributed

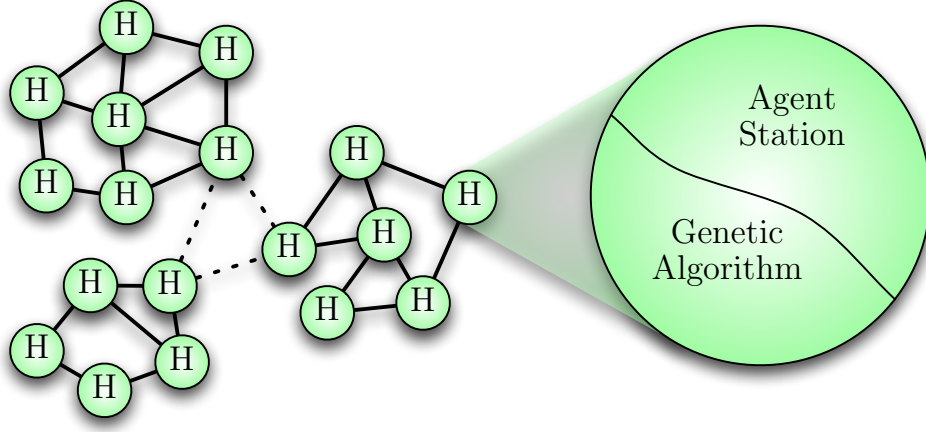


Figure 4: *Digital Ecosystem: Optimisation architecture in which agents (representing services) travel along the P2P connections; in every node (habitat) local optimisation is performed through an evolutionary algorithm, where the search space is determined by the agents present at the node.*

applications [36]. The Digital Ecosystem extends this concept with the automatic combining of available and applicable services, represented by agents, in a scalable architecture, to meet user requests for applications. These agents will recombine and evolve over time, constantly seeking to improve their effectiveness for the user base. From the SME users' point of view the Digital Ecosystem provides a network infrastructure where connected enterprises can advertise and search for services (real-world or software only), putting a particular emphasis on the composability of loosely coupled services and their optimisation to local and regional, needs and conditions. To support these SME users the Digital Ecosystem is satisfying the companies' business requirements by finding the most suitable services or combination of services (applications) available in the network. An application (composition of services) is defined as an agent aggregation (collection) in the habitat network that can move from one peer (company) to another, being hosted only in those where it is most useful in satisfying the SME users' business needs.

The agents consist of an *executable component* and an *ontological description*. So, the Digital Ecosystem can be considered a MAS which uses *distributed evolutionary computing* [12, 44] to combine suitable agents in order to meet user requests for applications.

The landscape, in energy-centric biological ecosystems, defines the connectivity between habitats [3]. Connectivity of nodes in the digital world is generally not defined by geography or spatial proximity, but by information or semantic proximity. For example, connectivity in a peer-to-peer network is based primarily on bandwidth and information content, and not geography. The island-models of distributed evolutionary computing use an information-centric model for the connectivity of nodes (*islands*) [28]. However, because it is generally defined for one-time use (to evolve a solution to one problem and then stop) it usually has a fixed connectivity between the nodes, and therefore a fixed topology [12]. So, supporting evolution in the Digital Ecosystem, with a multi-objective *selection pressure* (fitness landscape with many peaks), requires a re-configurable network topology, such that habitat connectivity can be dynamically adapted based on the observed migration paths of the agents between the users within the habitat network. Based on the island-models of distributed evolutionary computing [28], each connection between the habitats is bi-directional and there is a probability associated with moving in either direction across the

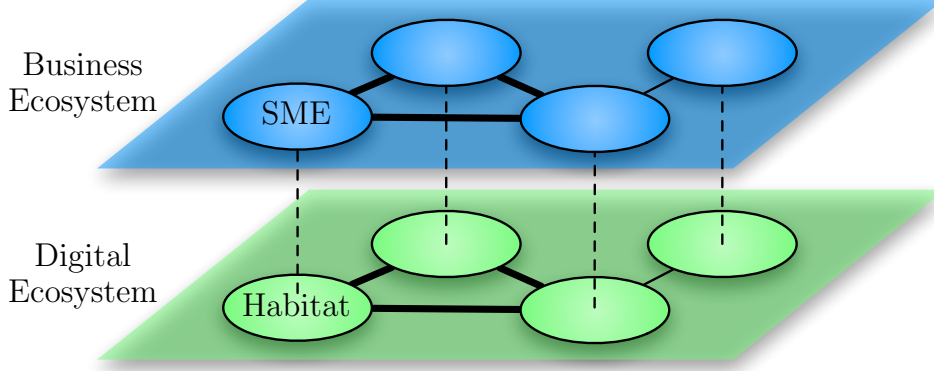


Figure 5: *Digital Business Ecosystem: Business ecosystem, network of SMEs, using the Digital Ecosystem. The habitat clustering will therefore be parallel to the business sector communities.*

connection, with the connection probabilities affecting the rate of migration of the agents. However, additionally, the connection probabilities will be updated by the success or failure of agent migration using the concept of Hebbian learning: the habitats which do not successfully exchange agents will become less strongly connected, and the habitats which do successfully exchange agents will achieve stronger connections. This leads to a topology that adapts over time, resulting in a network that supports and resembles the connectivity of the user base. If we consider a *business ecosystem*, network of SMEs, as an example user base; such business networks are typically small-world networks [47]. They have many strongly connected clusters (communities), called *sub-networks* (quasi-complete graphs), with a few connections between these clusters (communities). Graphs with this topology have a very high clustering coefficient and small characteristic path lengths [46]. So, the Digital Ecosystem will take on a topology similar to that of the user base, as shown in Figure 5.

The novelty of our approach comes from the evolving populations being created in response to *similar* requests. So whereas in the island-models of distributed evolutionary computing there are multiple evolving populations in response to one request [28], here there are multiple evolving populations in response to *similar* requests. In our Digital Ecosystems different requests are evaluated on separate *islands* (populations), and so adaptation is accelerated by the sharing of solutions between evolving populations (islands), because they are working to solve similar requests (problems).

The users will formulate queries to the Digital Ecosystem by creating a request as a *semantic description*, like those being used and developed in SOAs, specifying an application they desire and submitting it to their local peer (habitat). This description defines a metric for evaluating the *fitness* of a composition of agents, as a distance function between the *semantic description* of the request and the agents' *ontological descriptions*. A population is then instantiated in the user's habitat in response to the user's request, seeded from the agents available at their habitat. This allows the evolutionary optimisation to be accelerated in the following three ways: first, the habitat network provides a subset of the agents available globally, which is localised to the specific user it represents; second, making use of applications (agent aggregations) previously evolved in response to the user's earlier requests; and third, taking advantage of relevant applications evolved elsewhere in response to similar requests by other users. The population then proceeds to evolve the optimal application (agent aggregation) that fulfils the user request, and as the agents are the base unit for